

## Aiyesha Talat

Apex, North Carolina

aiyeshat@yahoo.com

### Professional Summary

Microsoft-certified **AI-102 Engineer** with 3.5 years of experience in **developing, training, deploying, and optimizing machine learning models**. Expertise in **natural language processing (NLP)**, **computer vision**, and **predictive analytics**. Proficient in **Azure Cognitive Services**, **data preprocessing workflows**, and deploying scalable AI models in cloud environments. Skilled in **cross-functional collaboration**, mentoring, and implementing cutting-edge AI solutions that adhere to **responsible AI practices**. Seeking to leverage technical expertise in creating innovative, consumer-centric AI/ML technologies.

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### Certifications

- **AI-102**: Designing and Implementing AI Solutions
  - **DP-100**: Data Science on Azure (In Progress)
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### Technical Skills

- **AI/ML Models**: NLP, Computer Vision, Predictive Analytics
  - **Cloud Platforms**: Azure, REST APIs, SDKs
  - **Programming**: Python
  - **Data Engineering**: Data Pipelines, Feature Engineering, Normalization
  - **Model Evaluation**: AUC-ROC, Precision, Recall, Cross-Validation, F1-Score
  - **Optimization**: Hyperparameter Tuning, Regularization, Ensemble Learning
  - **Production Deployment**: Scalable Cloud Systems
  - **Collaboration**: Cross-Functional Teams, Mentorship
  - **Documentation**: Model Reports, Pipeline Documentation
  - **Research**: Emerging AI/ML Techniques
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### Professional Experience

#### AI Engineer

Amnet Digitals(Intern) July 2025\_Present

## **Retrieval-Augmented Generation (RAG) Projects — Azure OpenAI, LangChain, FAISS, Azure AI Search**

- **Developed a Hybrid RAG pipeline integrating Azure OpenAI, LangChain, FAISS, and BM25** to perform both semantic and lexical document retrieval, significantly improving query precision.
- **Implemented OCR-based text extraction and caching using Azure Form Recognizer and Blob Storage**, reducing redundant API calls and enhancing document processing efficiency for large-scale PDF datasets.
- **Engineered a hybrid search algorithm combining BM25 keyword ranking with FAISS vector embeddings**, optimizing document retrieval through an adjustable alpha-weighted scoring mechanism.
- **Automated secure data handling with environment variables and Azure Blob Storage**, ensuring reproducibility and compliance with enterprise security standards.
- **Built and tested a Retrieval-Augmented Generation (RAG) demo integrating Azure OpenAI and Azure AI Search**, generating grounded, source-cited responses for enterprise Q&A use cases.
- **Implemented document filtering logic to extract unique, relevant chunks from Azure Search results**, improving retrieval accuracy and minimizing redundancy in knowledge base queries.
- **Integrated Azure AI Evaluators — DocumentRetrievalEvaluator, GroundednessEvaluator, and ResponseCompletenessEvaluator** — to assess factual grounding and completeness of LLM outputs, improving reliability.
- **Optimized LLM response accuracy and explainability by combining keyword-based retrieval with automated metric evaluation (nDCG, fidelity, groundedness score)** to ensure transparent performance benchmarking.
- **Developed and configured secure Azure clients using Azure Identity, SearchClient, and OpenAI SDK**, adhering to best practices in authentication and data protection.
- **Demonstrated end-to-end RAG functionality with formatted output and controlled grounding prompts**, showcasing LLM reasoning integrated with enterprise search.

## **Semantic Kernel Chat Completion — Azure OpenAI Integration**

- **Developed a chat completion pipeline using Azure OpenAI and Semantic Kernel**, enabling dynamic, real-time interaction through the AzureChatCompletion service and asynchronous API calls.
- **Configured and managed secure AI service connections with masked credentials and environment-safe setup practices**, ensuring reusable, compliant integration with Azure OpenAI deployments.

- **Implemented asynchronous prompt execution using PromptExecutionSettings and ChatHistory**, fine-tuning parameters such as temperature and token limits for optimal response control.
- **Documented and containerized the project setup with detailed README.md and requirements.txt**, improving reproducibility and onboarding for new developers learning Semantic Kernel integration.

## **Microsoft “Agents in Copilot Studio” – 5-Day Applied Workshop (*Self-directed Hands-On Training, Completed*)**

- **Independently explored building enterprise conversational agents using Microsoft Copilot Studio**
- **Integrated knowledge sources (Dataverse tables, PDFs, Office files) to support grounded and accurate responses**
- **Implemented tool-based actions to automate workflows such as data lookups and transactional updates**
- **Designed autonomous triggers and conditional logic to enable proactive agent behavior**
- **Tested and validated end-to-end AI agent interactions inside Microsoft 365 Copilot Chat**
- **Gained exposure to responsible AI controls, monitoring, and publishing lifecycle management**

**Reality AI Lab(Intern)Jan2025-Jun2025**

### **Presentation Generator**

- **Developed an AI-powered presentation generator** capable of creating slides with or without content based on user inputs such as instructional level, topic, language, and slide count, streamlining educational content creation.
- **Implemented an outline generation workflow using the Gemini-1.5-Flash model** and structured output via Pydantic, ensuring reliable mapping of content to slide-level structure.
- **Automated slide image creation using Imagen-3.0-Generate-002** for each outline item, delivering high-quality visuals aligned with the generated presentation content.
- **Integrated cloud storage and URL access** by storing generated images in Google Cloud Storage, enabling scalable, on-demand access for presentation rendering.

### **Image Generator**

- **Benchmarked multiple AI image generation tools** to evaluate performance, scalability, and reliability, optimizing for rapid image creation within milliseconds.

- **Designed a content safety validation pipeline** to filter prompts for violence, inappropriate content, discriminatory language, sensitive topics, and emotional triggers, ensuring ethical AI outputs.
- **Enhanced prompts using LangChain templates** to generate high-quality, contextually accurate images via Google's Imagen model, improving user satisfaction and visual relevance.
- **Automated image storage and retrieval** by saving outputs to Firebase or cloud storage buckets, enabling seamless access and integration into downstream applications.

## September 2024 – Jan 2025

- Some of my recent projects include generating Microsoft slide outlines and content and images ( generated using LLM).
  - Designed and deployed **machine learning models** for NLP (chatbot scheduling) and computer vision (image classification for fruits and dental images) using **Azure Cognitive Services**.
  - Preprocessed data and engineered features for robust **model training and evaluation**, ensuring data quality.
  - Deployed AI models in **production cloud environments**, optimizing for performance and scalability.
  - Conducted **model evaluation** using metrics like **AUC-ROC** and **F1-score**; applied **hyperparameter tuning** for improvement.
  - Collaborated with data scientists, software engineers, and stakeholders to define use cases and integrate solutions.
  - Documented ML workflows, pipelines, and best practices for internal and external audiences.
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## Workday Integration Specialist

### June 2023 – August 2023

- Streamlined **Workday HCM systems** by integrating AI-driven enhancements for seamless functionality and high data accuracy.
  - Developed **data pipelines** to ensure smooth workflows and optimized system performance.
  - Mentored team members and provided training on Workday HCM integrations and functionalities.
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## Educator

### January 2018 – June 2020

- Delivered structured lessons, honing **problem-solving and planning skills** applicable to AI project management.
  - Developed effective communication strategies for diverse audiences, fostering collaboration in technical projects.
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## Notable Projects

### Custom Vision Model – Fruit Classification

- **Task:** Developed a custom vision model to classify fruit images using Azure Cognitive Services.
- **Outcome:** Achieved >95% accuracy with real-time SDK integration.

### Custom Vision Model – Dental Image Analysis

- **Task:** Built a classification model to identify infection patterns in dental images.
- **Outcome:** Successfully deployed a scalable cloud solution with 92% confidence scores.

### NLU Chatbot for Appointment Scheduling

- **Task:** Designed and implemented an NLP-based chatbot for healthcare appointment scheduling.
  - **Outcome:** Improved scheduling efficiency and user satisfaction with AI-driven automation.
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**Work Authorization:** US Citizen